

INTERNAL DEVIATION & NON-CONFORMANCE NOTIFICATION

Deviation Number:	DEV-2024-0814-A01	Date of Occurrence:	14-08-2024
Site / Plant:	API Manufacturing Unit - Block B	Source of Event:	Manufacturing (In-Process Control)
Product / Material:	Metformin Hydrochloride API (BP/USP)	Batch / Lot Number:	MFH260712A
Equipment Involved:	Glass-Lined Reactor R-102	Initial Severity:	High (Process Parameter OOS)

1. Detailed Incident Description:

During Stage-II crystallization of Metformin Hydrochloride synthesis in Glass-Lined Reactor R-102 (Batch No: MFH260712A), the automated chilled brine supply valve (CV-102B) malfunctioned and became locked in the 15% open position. This caused the internal crystallization temperature to rise from the validated target range of 20.0°C – 24.0°C to an excursion peak of 48.5°C. The out-of-specification temperature condition persisted for approximately 42 minutes before the plant operator noticed the local alarm and initiated manual bypass override. The crystallization process was subsequently completed under manual cooling control.

2. Immediate Containment Actions Taken:

- Batch MFH260712A was immediately quarantined in the centrifuge holding area.
- A physical red 'QA HOLD' tag was affixed to centrifuge C-02 and mother liquor filtrate receivers.
- Instrumentation maintenance was notified; valve positioner calibration confirmed mechanical jamming.
- Composite samples were sent to the QC Laboratory for accelerated impurity profiling and crystal polymorph testing.

3. Quality Assurance Initial Impact Assessment:

Initial impact is classified as Major. Thermal excursion above 40°C during crystallization may promote unwanted degradation byproducts (specifically Related Substance Impurity-A / cyanoguanidine) and potentially alter the particle size distribution (PSD) and powder bulk density. Batch release is blocked pending full chromatographic purity re-testing and root cause investigation.